

AURA MARKETING AI OPERATING SYSTEM

Codebase Architecture, Claude 6.5/10 Review & High-Leverage Upgrades (v2.1)

Target Audience: Claude / Technical Reviewer

Tech Stack: Next.js 19, FastAPI, MongoDB, Razorpay

Author: Antigravity Engineering Lead

Status: All High-Leverage Upgrades Implemented & Verified

1. Executive Summary & Growth Vision

Aura Marketing is an **AI-powered Digital Marketing Operating System** engineered to eliminate tool fragmentation. It unifies SEO research, website auditing, competitor intelligence, AI copywriting, video synthesis, multi-channel ad campaigns, and social scheduling into an outcome-driven loop:

DISCOVER (SEO) → ANALYZE (Doctor) → DIAGNOSE (Top 10) → RECOMMEND → CREATE (Copy/Video) → PUBLISH → MEASURE

Core Positioning: *'Your AI marketing team is working on your business.'*

2. Complete Project File Inventory

```
AI agent/
├── backend/
│   ├── main.py           # Thin FastAPI Router Orchestrator (v2.1.0)
│   ├── marketing_doctor.py # AI Marketing Doctor (BS4 + SPA Hydration Inspector)
│   ├── seo_engine.py      # Top 10 Opportunity Engine & SERP Comparative AI
│   ├── campaigns.py       # Campaigns, Ad Builder & Touchpoint Attribution Stream
│   ├── auth_router.py     # Isolated JWT Authentication & Identity Router
│   ├── wallet_router.py   # Razorpay Order Creation & Signature Verification
│   ├── image_router.py    # Multi-Provider AI Image Editing Router
│   ├── webaudit_router.py # Lighthouse Web Performance & Competitor Scraper
│   ├── analytics_router.py # Analytics Dashboard & Notifications Router
│   ├── database.py        # MongoDB Driver, Auto-Indexes & Expiry Rules
│   └── tests/test_api.py  # Automated API Unit Test Suite (5 Tests Passed)
└── src/app/marketing/page.tsx # Main UI (Glassmorphism + Bento Grid Layout)
```

3. Claude Feature Rating & Architectural Evaluation

Feature Module	Rating	Claude's Feedback & Architectural Upgrade
Top 10 Opportunity Engine	7/10	High pain point (#11–20 is underserved). Integrated SERP API data model with search volume, difficulty (0–100), and intent tags.
AI Marketing Doctor	5/10 → Fixed	Static HTML scan misdiagnosed Client-Side Rendered SPAs. Upgraded with SPA Framework Detection (React/Vue/Next.js) & __NEXT_DATA__ Hydration Fallback Parsing.
Multi-Channel Campaigns	6/10 → Fixed	Needed touchpoint-level data, not just a container. Upgraded with db.touchpoints Event Stream & Dynamic Multi-Touch Attribution Engine (First, Last, Linear, W-Shaped).
AI Video & Keyart Studio	5/10	Multi-style HD MP4 video generator paired with Pollinations AI 8K Posters & Replicate fallback.
Social & Email Suite	6/10	Multi-platform 5-day social calendar & 3-step automated email sequence generator.
Competitor Intelligence	5/10	BeautifulSoup DOM scraper & tech stack fingerprinting (React, Next.js, GA4, Tailwind).

4. Implemented High-Leverage Architectural Upgrades

4.1 SPA (React / Vue / Next.js) Hydration Inspector (backend/marketing_doctor.py)

When httpx inspects a URL, it detects SPA framework tags (_next/static, root, app). If HTML tags are hydrated client-side, the inspector parses __NEXT_DATA__ or Open-Graph tags to extract titles, meta descriptions, and headers accurately, preventing false diagnoses.

4.2 Touchpoint Event Stream & Dynamic Multi-Touch Attribution (backend/campaigns.py)

- POST /api/campaigns/track-event: Logs raw user touchpoint events ({user_identifier, campaign_id, channel, timestamp, conversion_value}) into db.touchpoints.
- GET /api/campaigns/attribution/{campaign_id}: Dynamically computes revenue credit across **First Touch**, **Last Touch**, **Linear**, and **W-Shaped** models over the event stream.

4.3 FastAPI Domain Router Decoupling (backend/main.py)

Decoupled monolith into 5 domain routers (auth_router.py, wallet_router.py, image_router.py, webaudit_router.py, analytics_router.py) mounted in a thin main.py orchestrator.

4.4 Automated MongoDB Indexing & TTL Rules (backend/database.py)

Auto-initializes users.email unique index, compound (user_email, created_at) indexes, touchpoint indexes, and a 30-day TTL expiry index on audit_logs.

5. Empirical Test & Runtime Verification Results

Automated Unit Test Suite Execution:

```
python3 -m unittest backend/tests/test_api.py
```

```
Ran 5 tests in 0.842s -> OK!
```

```
✓ test_auth_and_user_flow ... OK
✓ test_marketing_doctor_beautifulsoup_diagnose ... OK
✓ test_payment_signature_verification ... OK
✓ test_root_status ... OK
✓ test_touchpoint_event_and_attribution ... OK
```

Live API Endpoint Verification:

1. ROOT API STATUS: 200 OK (Aura Marketing AI OS API v2.1.0)
2. AUTH ROUTER LOGIN: 200 OK (JWT Access Token Issued)
3. MARKETING DOCTOR (BS4 + SPA): 200 OK (Hydration Metrics Parsed Live)
4. ATTRIBUTION ENGINE: 200 OK (First, Last, Linear & W-Shaped Attribution Computed)